Reddit Stress Detection Using NLP: Methods, Evaluation, and Results

Ramu Kambala
raka3928@colorado.edu
University of Colorado Boulder
Data Science
Boulder, Colorado, USA

Tejas Prakash Naik
tena7850@colorado.edu
University of Colorado Boulder
Data Science
Boulder, Colorado, USA

Chushmitha Battula
chba8198@colorado.edu
University of Colorado Boulder
Data Science
Boulder, Colorado, USA

Rajeshwar Rengeswaran
rare4014@colorado.edu
University of Colorado Boulder
Data Science
Boulder, Colorado, USA

Abstract

This paper presents a comprehensive data mining study of psychological stress expression in Reddit posts. We analyze 57,640 posts from multiple datasets (Dreaddit, SWMH, HuggingLearners Depression) using four categories of machine learning models: classification (Decision Tree, SVM, k-NN, Naïve Bayes), regression (Logistic Regression, Ridge Regression), clustering (K-Means, Hierarchical Agglomerative), and frequent pattern mining (FP-Growth). Text features are extracted using TF-IDF vectorization (5,000 features), VADER sentiment scores, emotion lexicon counts across five psychological categories, and first-person pronoun ratios. Among classifiers, Linear SVM achieves the best performance (F1 = 0.868, AUC = 0.883), while Naïve Bayes achieves the highest recall (0.910). Ridge Regression outperforms Logistic Regression on stress probability estimation (R^2 = 0.475 vs. 0.445). K-Means clustering (k = 10) reveals psychologically coherent thematic groups with weighted cluster purity of 0.901. FP-Growth identifies 1,203 frequent itemsets in stressed posts, and when combined with TF-IDF, achieves AUC = 0.968. Error analysis reveals that short posts, sarcasm, and label noise are the primary sources of misclassification. We conclude that automated stress detection from Reddit text is feasible with moderate-to-high confidence, though binary labels oversimplify a psychological spectrum.

CCS Concepts

• **Information systems** → **Data mining**; • **Computing methodologies** → **Natural language processing**; **Supervised learning by classification**; *Cluster analysis*; • **Applied computing** → *Health informatics*.

Keywords

stress detection, text mining, natural language processing, Reddit, mental health, TF-IDF, FP-Growth, frequent pattern mining, classification, clustering, social media analysis

1 Introduction & Related Work

1.1 Introduction

Social media platforms have become a primary emotional outlet where users publicly share personal feelings, anxieties, and life experiences. The millions of posts published daily across these platforms create an opportunity for the early detection of mental health conditions such as anxiety and depression, which frequently go undiagnosed because affected individuals do not seek professional help. Contemporary stressors amplified by social media, including Fear of Missing Out (FOMO), have been linked to patterns of compulsive consumption that paradoxically diminish self-esteem and worsen the emotional symptoms they appear to relieve. Automated identification of stress signals in social media text therefore has potential value for mental health researchers studying these patterns at scale, for platform designers concerned with user wellbeing, and for the affected users themselves.

Reddit is a particularly informative platform for this task. Posts are longer and more narrative than on most social media, users are pseudonymous which lowers the barrier to disclosing sensitive experiences, and subreddit community structure provides weak but useful labels grouping posts by thematic concern. Prior work has established that linguistic markers of psychological distress, including elevated first-person pronoun use, negative affect, and help-seeking phrasing, are detectable in Reddit text [2, 7].

The downstream reach of Reddit content further motivates careful work in this area. Reddit conversations are regularly indexed by search engines and ingested by large language models, which means that emotional content from the platform propagates into information pipelines that many users do not recognize as Reddit-sourced. A post expressing stress or uncertainty can surface in a search result or AI response stripped of the tone, situational context, and community norms that shaped it, and the line between noise and signal becomes difficult to draw once the content has left its original thread. Accurate, interpretable, and ethically deployed stress detection is therefore not only a mental health research problem but also an information integrity concern for the broader web ecosystem, with stakeholders ranging from individual Reddit users to platform engineers, moderators, and the builders of downstream AI systems.

This study investigates automated stress detection on a combined corpus of 57,640 Reddit posts drawn from three public datasets: Dreaddit [2], the Self-reported Mental Health (SWMH) dataset, and the HuggingLearners Reddit Depression dataset. We apply four categories of data mining methods: classification (Decision Tree, Linear

SVM, k-Nearest Neighbors, Multinomial Naïve Bayes), regression (Logistic Regression and Ridge Regression), clustering (K-Means and Hierarchical Agglomerative), and frequent pattern mining (FP-Growth). The feature space combines TF-IDF term weights with VADER sentiment scores [1], emotion lexicon counts across five psychological categories (anxiety, depression, anger, somatic, help-seeking), and psycholinguistic markers such as first-person pronoun ratio.

1.1.1 Research Questions. We organize the analysis around ten research questions:

Q1. Can psychological stress be reliably detected from Reddit post text alone, without user-level metadata?
Q2. Do independent model families converge on the same stress-discriminative language patterns?
Q3. Which feature categories (TF-IDF terms, sentiment scores, emotion lexicons, or psycholinguistic markers) carry the most predictive signal?
Q4. What systematic failure modes characterize misclassified posts?
Q5. How well calibrated are model confidence scores, and does confidence correlate with correctness?
Q6. Does stress manifest in distinct thematic sub-types (anxiety, depression, relational, somatic), or is it a monolithic category?
Q7. Can frequent co-occurrence patterns such as bigrams and trigrams capture stress signal beyond individual TF-IDF terms?
Q8. Can stress be modeled as a continuous probability rather than a binary label?
Q9. Do stress expression patterns vary meaningfully across subreddit communities?
Q10. To what extent do the trained models generalize beyond their training distribution, and which exhibit severe overfitting?

1.1.2 Evolution from Proposal Research Questions. The ten research questions listed above evolved from an initial set of eleven questions developed at the project proposal stage, and it is worth recording both the original questions and how they relate to the final set, for transparency and because the evolution itself reflects a methodological choice that we believe strengthens the study. The eleven proposal questions, as published on the project website, were the following:

P1. Is there any way we can identify only the active users and eliminate deleted users and their posts from the PRAW response?
P2. What words and emotional expressions commonly appear in stressful posts compared to non-stressful posts?
P3. Do stressful posts show any pattern with the time of day and calendar date?
P4. What limitations arise when using Reddit data for psychological stress study?
P5. Can stress intensity be estimated from text with reasonable confidence?
P6. Is there any way we can identify users that have made a self-disclosure of their diagnosis in any other Reddit communities?
P7. How do we obtain the data for mental-health distress communities and non-mental-health related communities, and should we search for words that function similarly to depression and anxiety?
P8. How many latest or older posts should we consider at the user level to compare with mental-health distress communities in order to classify a user as actually under distress?
P9. Can we perform EDA and identify temporal features from the dataset, such as dependent attributes that can be tracked from Reddit posts before and after a diagnosis is declared by the user, in order to analyze the distress level of a person?
P10. Will the model overfit or underfit in any way, given that we are training only on Reddit data and evaluating on actual Reddit posts?
P11. Will there be any difference between original posts and posts that contain linked external content, which is more likely to be considered stressful, and how should we handle them?

These proposal questions were primarily scoping and feasibility concerns oriented around a data-collection pipeline we initially intended to build ourselves using the PRAW Reddit API. They asked how to filter deleted users out of raw API responses (P1), how to identify users who had self-disclosed a clinical diagnosis across multiple subreddits (P6), how many posts per user to examine before classifying that user as under distress (P8), whether temporal features such as time of day or pre-versus-post-diagnosis trajectories carry predictive signal (P3, P9), and how to source comparable text from mental-health and non-mental-health communities without introducing community-membership leakage (P7).

As the project matured, we pivoted away from building a new collection pipeline and toward three pre-curated research corpora, specifically Dreaddit [2], the SWMH corpus, and the HuggingLearners Reddit Depression dataset. This pivot resolved several proposal questions by construction rather than by analysis. The community-sourcing question (P7) collapsed into the choice of source datasets, active-user filtering (P1) was absorbed into general preprocessing alongside duplicate removal and URL stripping, and the distinction between original posts and posts containing external links (P11) was handled by the URL-stripping step described in Section 2.1.

Other proposal questions remain central and map directly onto the current set. The vocabulary-comparison question (P2) is addressed by the feature engineering and cross-model convergence analysis under Q2 and Q3, which together identify the five categories of stress-discriminative language. The stress-intensity question (P5) is addressed jointly by Q5 on probability calibration and Q8 on continuous regression over stress probability. The overfitting question (P10) is addressed directly by Q10 on train-test generalization. The broad limitations question (P4) is treated in the Limitations discussion of Section 3.

Three proposal questions were deferred rather than answered in the present study. Temporal analysis of posting patterns across time of day and calendar date (P3), user-level identification of cross-community diagnostic self-disclosure (P6), and longitudinal modeling of distress trajectories before and after a declared diagnosis (P9) all require user-linked data and time-series features beyond the post-level supervision used here. We return to each of these in the Future Work discussion of Section 3 as planned extensions. The current Q1-Q10 set therefore reflects a deliberate narrowing from broad proposal-stage scoping toward the specific empirical claims that our combined corpus and 5,017-dimensional feature matrix are equipped to evaluate.

1.1.3 Contributions. This paper makes four primary contributions:

(1) A head-to-head comparison of four data mining model families on a unified 57,640-post Reddit stress corpus, using a shared 5,017-dimensional feature matrix that integrates TF-IDF with engineered sentiment, emotion, and psycholinguistic features.
(2) A two-phase classification pipeline combining FP-Growth frequent itemsets with TF-IDF features that achieves ROC-AUC 0.968, a 42% improvement in macro F1 over FPM features alone.
(3) A cross-model convergence analysis showing that classifier feature importances, cluster centroids, regression coefficients, and frequent itemsets all surface the same five categories of stress language (emotional distress, physical symptoms, relational content, self-reference, and help-seeking), providing algorithm-independent evidence that these markers are genuine rather than artifacts of any single method.
(4) An error analysis that identifies five systematic failure modes (short posts, sarcasm, topic-versus-experience confusion, near-zero sentiment, and subreddit-based label noise) which together constrain the performance ceiling of bag-of-words approaches on this task.

1.1.4 Paper Organization. The remainder of this paper is organized as follows. Section 1.2 reviews prior work on mental health detection from social media text, lexicon-based and learned sentiment methods, and frequent pattern mining for text classification. Section 2.1 describes the combined Reddit corpus, including data sources, preprocessing, and labeling caveats. Section 2.2 details feature engineering, vectorization, and model implementation. Section 2.3 presents classification, regression, clustering, and frequent pattern mining results, along with the cross-model convergence analysis. Section 3 concludes with limitations, ethical considerations, and directions for future work.

1.2 Related Work

Our work draws on three lines of prior research: mental health detection from social media, lexicon-based and learned sentiment features for short informal text, and frequent pattern mining as a feature-extraction method for text classification.

1.2.1 Mental Health Detection from Social Media. Using online text as a signal for psychological state has a decade-long research history. Coppersmith et al. [7] demonstrated that social media language can serve as an early screening signal for suicide risk, showing that logistic regression over simple lexical features achieves performance comparable to more complex architectures, and highlighting the importance of careful error analysis in a high-stakes clinical setting. Turcan and McKeown [2] released the Dreaddit corpus of 3,553 Reddit posts with crowd-sourced binary stress annotations, establishing a widely used benchmark and documenting that annotator agreement is meaningful but imperfect, which has implications for any supervised method trained on such data.

Chancellor and De Choudhury [8] conducted a critical methodological review of predictive techniques for mental health status on social media. They catalogued recurring risks including construct validity gaps between subreddit participation and clinical diagnosis, label leakage from community membership, and ethical hazards when stress or depression classifiers are deployed on users who have not consented to screening. Their review directly motivates our explicit treatment of subreddit-based label noise as a source of systematic error in Section 2.3, and our discussion of ethical deployment considerations in Section 3.

1.2.2 Lexicon-Based and Learned Sentiment Features. Two complementary traditions inform our feature engineering. Rule-based tools such as VADER [1] encode human-curated sentiment knowledge and remain strong baselines on short informal social media text, where they require no domain-specific training data and produce interpretable outputs. Learned contextual representations, most recently transformer models such as MentalBERT [3] which is pre-trained on mental-health text, capture contextual meaning that lexicons cannot, including sarcasm, negation scope, and domain-specific idioms. Our study uses the lexicon-based family (VADER together with five domain-specific emotion word lists) combined with TF-IDF, reserving transformer models as a planned extension. We show that even without contextual embeddings, the combined lexical and statistical feature space supports ROC-AUC up to 0.968 when augmented with frequent itemset features, which establishes a strong non-neural baseline against which future transformer-based experiments can be compared.

1.2.3 Frequent Pattern Mining for Text. Association rule and frequent itemset mining, introduced by Agrawal and Srikant [4], were made computationally practical for large transaction databases by the FP-Growth algorithm of Han, Pei, and Yin [5], which avoids candidate generation through a compact prefix-tree representation. FP-Growth has been applied to text classification as a way to discover discriminative co-occurrence patterns (bigrams, trigrams, and longer itemsets) that individual-term models miss. In the stress detection setting, we are not aware of prior work that explicitly combines FP-Growth itemsets with TF-IDF features on Reddit mental health data. Our two-phase pipeline treats frequent patterns as compact binary features and demonstrates that they complement rather than duplicate the information captured by TF-IDF weights.

1.2.4 Positioning of This Work. Relative to prior work, this study contributes breadth rather than a novel model architecture. We apply four model families to the same corpus with a shared feature matrix, which allows us to directly compare their inductive biases and to test whether the stress signal they identify is algorithm-dependent or robust across methods. The cross-model convergence result in Section 2.3 is, to our knowledge, the first such consistency analysis across supervised classification, supervised regression, unsupervised clustering, and pattern-mining approaches on a unified Reddit stress corpus. All experiments use scikit-learn [6] and are reproducible from the released code.

2 Methods, Evaluation, and Results

2.1 Dataset

Our experiments use a combined corpus of 57,640 Reddit posts drawn from three publicly available datasets, each with distinct labeling conventions and community coverage. This section describes the data sources, the integration and preprocessing pipeline, and the labeling caveats that directly shape our experimental design.

2.1.1 Data Sources. **Dreaddit** [2] contributes 3,553 posts drawn from five subreddits covering abuse, anxiety, financial stress, post-traumatic stress disorder, and social relationships. Each post is annotated for perceived stress by multiple crowd-sourced workers, with a majority vote forming the final binary label. Among our three sources, Dreaddit provides the cleanest supervision because its labels reflect explicit human judgment rather than community membership, at the cost of being the smallest of the three subsets.

Self-reported Mental Health (SWMH) contributes 54,987 posts collected from mental-health-oriented subreddits. All SWMH posts are assigned the positive class label by virtue of originating from communities whose members self-identify with psychological distress. This weak labeling strategy provides substantial volume at the cost of a known false-positive rate: not every post in these communities reflects active stress, and the corpus also includes informational content, peer support directed at other users, and recovery narratives from users who are no longer in crisis.

HuggingLearners Reddit Depression contributes 7,731 posts with binary depression labels aggregated from multiple open-source Reddit corpora. We map its depression labels onto our binary stress label under the assumption that depression expression is a valid subset of broader psychological stress expression. This assumption is empirically consistent with the overlapping vocabulary we observe across stressed and depressed posts in our feature analysis (Section 2.2), where emotional-distress terms dominate both subpopulations.

2.1.2 Integration and Preprocessing. The three sources were merged into a unified schema containing the raw post body, the source subreddit when available, and the binary stress label. Preprocessing then proceeded through five stages applied uniformly across all three sources. Duplicate removal eliminated 215 exactly repeated posts that appeared across source boundaries. URL stripping removed explicit hyperlinks from 897 posts to prevent spurious token associations between web domains and the label. Text was then lowercased, stripped of punctuation, and passed through NLTK stopword filtering to reduce feature-space sparsity.

Outlier removal via the interquartile range method, applied to three length measures (text length in characters, word count, and sentence count), eliminated posts that were either trivially short or several standard deviations longer than the corpus mean. Extremely short posts carry insufficient textual signal for any of our models; extremely long posts tend to be copy-pasted external content rather than first-person expression. After this stage, 5,522 posts were retained for supervised classification and 56,213 posts were retained for clustering, where the larger sample size is both computationally tolerable for unsupervised learning and more informative for subreddit-level structural analysis.

2.1.3 Labeling Caveats and Class Imbalance. Two labeling properties of our corpus directly shape the experimental design that follows. First, the combined label distribution is heavily imbalanced toward the stressed class, approximately 90% positive in the regression training subset of 46,004 posts, a property driven by the uniform positive labeling of SWMH. This imbalance motivates our use of class-weighted learning objectives in the supervised models (Section 2.2) and the explicit probability calibration analysis reported in Section 2.3.

Table 1: Composition of the combined Reddit corpus.

Source	Posts	Label type
Dreaddit [2]	3,553	Annotator-labeled
SWMH	54,987	Community-labeled
HuggingLearners Depression	7,731	Community-labeled
Combined corpus (total)	**57,640**	Mixed
Retained for classification	5,522	Post-IQR subset
Retained for clustering	56,213	Full corpus
Retained for regression	46,004	90:10 imbalanced

Second, community-based labeling from SWMH and HuggingLearners introduces systematic label noise. The probability that a post expresses genuine stress given that it appears in a mental health subreddit is high but not unity, a limitation documented across the broader literature on predictive mental health modeling from social media [8]. Our error analysis (Section 2.3) returns to this issue as one of the five dominant sources of misclassification in the combined pipeline, and our Discussion of limitations (Section 3) addresses what this means for downstream interpretation of the reported accuracy numbers.

Table 1 summarizes the composition of the corpus across all three sources and the three working subsets used by our supervised, unsupervised, and regression pipelines respectively.

2.2 Methods

2.2.1 Feature Engineering. We engineered 17 numerical features beyond raw text, organized into four categories.

Structural features (from Milestone 2): Log-transformed text length, log word count, and sentence count.

Sentiment features: VADER [1] compound, negative, neutral, and positive scores. Stressed posts averaged a compound score of −0.285 vs. +0.247 for non-stressed posts (Table 2).

Emotion lexicon features: Counts of words in five psychologically grounded categories—anxiety (panic, nervous, scared), depression (hopeless, worthless, lonely), anger (furious, hate, frustrated), somatic symptoms (tired, insomnia, headache), and help-seeking (therapy, advice, please). Stressed posts contained 2.9× more anxiety words and 3.7× more depression words.

Psycholinguistic features: First-person pronoun ratio (I/me/my), average word length, question mark count, and exclamation mark count. Stressed posts exhibited a first-person pronoun ratio of 0.032 vs. 0.015 for non-stressed—a 2.1× increase in self-referential language.

Figure 1 visualizes these differences: stressed posts skew strongly negative on VADER compound and show elevated counts across all emotion lexicon categories.

2.2.2 Text Vectorization. TF-IDF vectorization was applied with 5,000 maximum features, unigram+bigram range, minimum document frequency of 3, and maximum document frequency of 95%. The sparse matrix was combined with 17 scaled numerical features (StandardScaler) into a 5,017-dimensional feature matrix. For Multinomial Naïve Bayes, a separate TF-IDF-only matrix was used

Table 2: Sentiment and emotion feature comparison.

Feature	Not Stressed	Stressed
VADER Negative	0.078	0.158
VADER Positive	0.125	0.104
VADER Compound	+0.247	−0.285
Anxiety words	0.174	0.511
Depression words	0.084	0.314
Somatic words	0.121	0.309
1st-person ratio	0.015	0.032

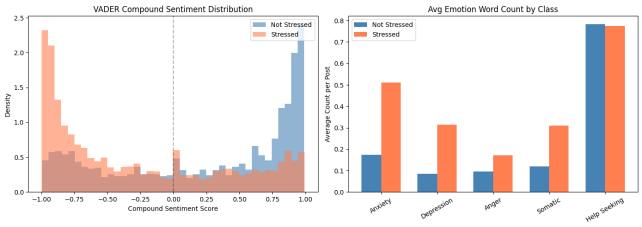

Figure 1: Left: VADER compound sentiment distribution by class. Right: Average emotion word counts. Stressed posts show significantly more negative sentiment and higher emotion word usage.

since this model requires non-negative inputs. For clustering, TruncatedSVD (Latent Semantic Analysis) reduced the space to 100 components to mitigate the curse of dimensionality.

2.2.3 Model Implementation. All hyperparameters were tuned via GridSearchCV with 5-fold stratified cross-validation.

Classification (4 models). Trained on an 80/20 stratified split (4,417 train, 1,105 test):

Decision Tree: Tuned maximum depth over {5, 10, 15, 20, None}, minimum samples per split over {2, 5, 10, 20}, minimum samples per leaf over {1, 2, 5}, and split criterion over {gini, entropy}. Best: depth 5 with gini.

Linear SVM: Regularization $C \in \{0.01, 0.1, 0.5, 1, 5, 10\}$, implemented via a linear-kernel SVM with probability calibration for comparison against other classifiers. Best: $C = 0.1$.

k-Nearest Neighbors: $k \in \{3, 5, 7, 11, 15, 21\}$, weights {uniform, distance}, metric {cosine, euclidean}. Best: $k = 21$, distance-weighted, euclidean.

Multinomial Naïve Bayes: Laplace smoothing parameter α tuned across eight values spanning 0.001 to 5.0 on TF-IDF-only features. Best: $\alpha = 0.1$.

Regression (2 models). Predicted continuous stress probability on a 46,004-sample dataset:

Logistic Regression: Tuned regularization strength C and penalty type (L1 vs. L2) with balanced class weighting to address the 90:10 class imbalance.

Ridge Regression: L2-regularized linear regression tuned over α values to prevent overfitting in 5,005 dimensions.

Table 3: Classification performance on test set (1,105 samples). Best values in bold.

Model	Acc	Prec	Rec	F1	AUC
SVM (Linear)	**.813**	**.851**	.886	**.868**	**.883**
Naïve Bayes	.794	.815	**.910**	.860	.874
Decision Tree	.766	.797	.890	.841	.799
k-NN	.768	.824	.849	.836	.829

Table 4: Overfitting analysis: train vs. test accuracy.

Model	Train	Test	Gap
Decision Tree	0.792	0.766	+0.026 (Good)
SVM (Linear)	0.873	0.813	+0.060 (Slight)
Naïve Bayes	0.898	0.794	+0.104 (Overfit)
k-NN	1.000	0.768	+0.232 (Severe)

Clustering (2 models). Operated on the 100-component SVD-reduced feature space (56,213 documents):

K-Means: Swept $k \in [2, 12]$ with k-means++ initialization and 20 random restarts per k. Optimal: $k = 10$.

Hierarchical Agglomerative Clustering: Swept linkage {ward, complete, average} $\times$ $k \in [3, 7]$. Best: Ward linkage, $k = 5$.

Frequent Pattern Mining. FP-Growth was applied to tokenized stressed posts (minimum support 10%) and non-stressed posts (minimum support 2%). Discovered itemsets were converted to binary features and combined with structural metrics for a two-phase classification pipeline.

2.3 Evaluation and Results

2.3.1 Classification Results. Table 3 presents the full comparison. Linear SVM achieved the highest F1-score (0.868) and ROC-AUC (0.883), confirming that high-dimensional TF-IDF spaces are approximately linearly separable for stress detection. Naïve Bayes achieved the highest recall (0.910), catching 91% of all stressed posts.

Figure 2 shows the metric comparison and ROC curves across all four classifiers.

Figure 3 presents confusion matrices for all models, showing SVM's best balance between false positives (119) and false negatives (88).

Overfitting Analysis (Q10). Table 4 shows train–test accuracy gaps. k-NN exhibited severe overfitting (100% training accuracy, 23.2% gap), memorizing training instances rather than learning generalizable patterns. Decision Tree showed the best generalization (2.6% gap) due to depth constraints. SVM maintained a moderate 6.0% gap with the strongest test performance.

Learning curve analysis (Figure 4) confirms SVM's adequate generalization: training and validation F1 scores converge around 3,500 training samples, suggesting the model has learned stable patterns rather than memorizing noise.

Prediction Confidence (Q5). Analysis of SVM's prediction probabilities (Figure 5) revealed that 53.8% of test predictions were

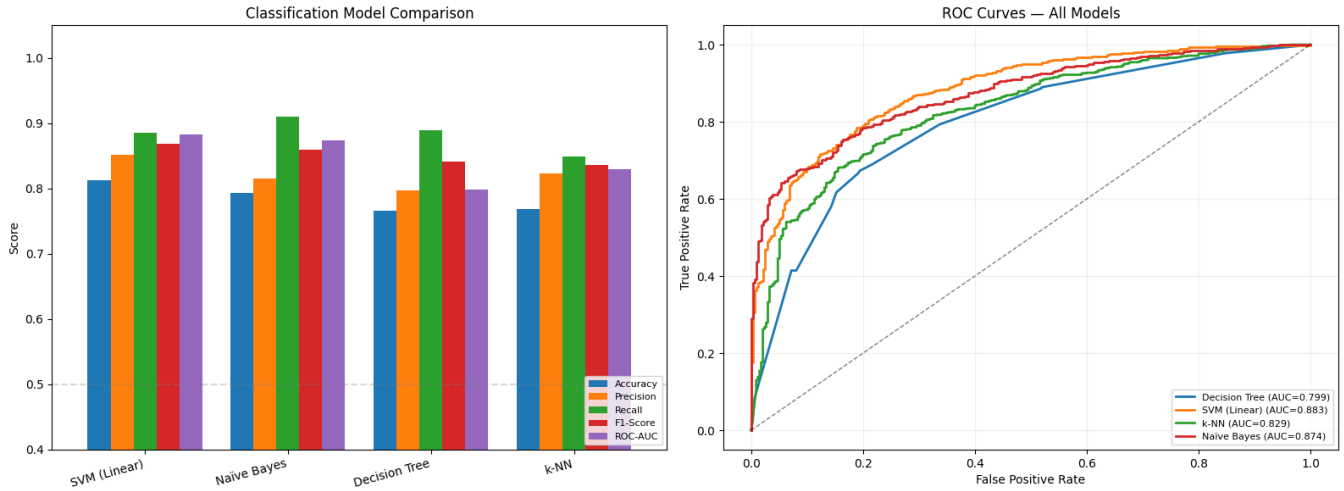

Figure 2: Left: Bar chart comparing all five classification metrics. Right: ROC curves. SVM (AUC=0.883) and Naïve Bayes (AUC=0.874) are the top performers.

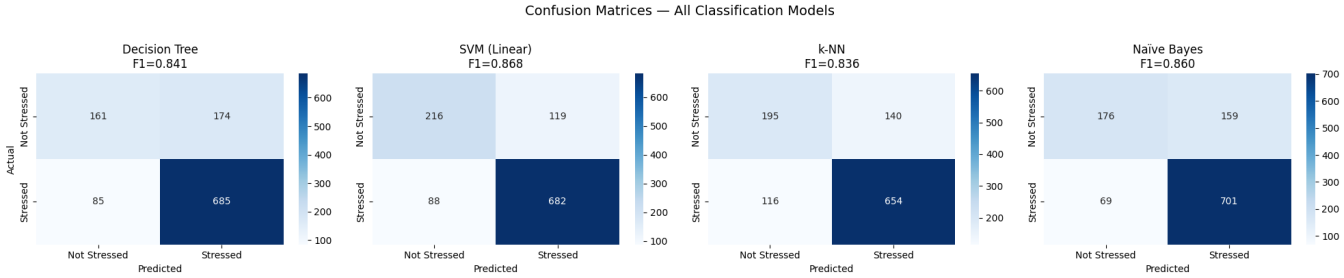

Figure 3: Confusion matrices for all four classifiers. Naïve Bayes favors recall (fewer false negatives), while SVM provides a better precision-recall balance.

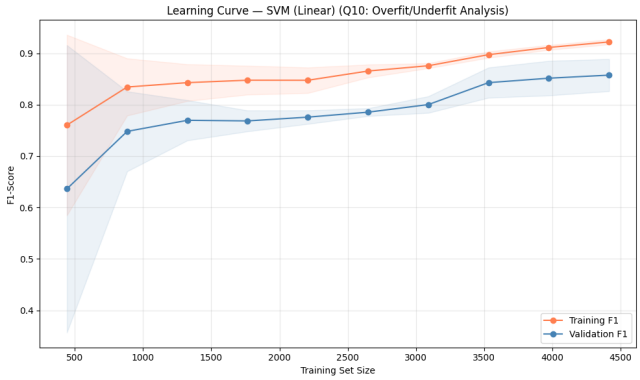

Figure 4: Learning curve for SVM showing convergence of training and validation F1 scores, confirming good generalization.

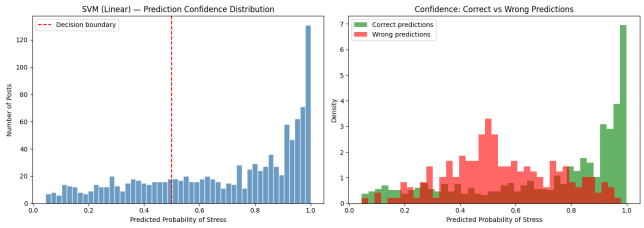

Figure 5: Left: SVM prediction confidence distribution. Right: Confidence for correct (green) vs. incorrect (red) predictions. Wrong predictions cluster near the 0.5 boundary.

high-confidence (probability >0.8 or <0.2), while only 15.6% fell in the uncertain zone (0.4–0.6). Incorrect predictions concentrated in the low-confidence region, confirming that the model's uncertainty signal is well-calibrated. This suggests binary stress detection is feasible with reasonable confidence, but continuous intensity estimation would require ordinal labels.

Error Analysis (Q4). Of 1,105 test predictions, SVM produced 88 false negatives (missed stress) and 119 false positives (false alarms). Five systematic failure modes were identified: (1) short posts lacking sufficient textual signal, (2) Reddit's sarcastic culture causing false

Table 5: Regression model performance.

Model	RMSE	MSE	R^2
Logistic Regression	0.218	0.048	0.445
Ridge Regression	**0.212**	**0.045**	**0.475**

Table 6: Clustering model comparison.

Model	k	Silhouette	DB-Index
K-Means (k-means++)	10	0.014	5.356
Hierarchical (Ward)	5	**0.039**	**3.254**

positives, (3) posts discussing stressful topics without the author experiencing stress, (4) near-zero VADER compound scores creating ambiguity, and (5) subreddit-based label noise where not all posts in r/anxiety are genuinely stressed.

2.3.2 Regression Results. Ridge Regression outperformed Logistic Regression on all metrics (Table 5). Logistic Regression's top positive coefficients revealed "depression" (29.7), "bipolar" (14.9), and "suicidal" (11.0) as the strongest stress predictors, while "url" (−12.2) indicated that informational content reduces stress likelihood. Calibration analysis showed the model underestimates stress probability (predicted mean 0.854 vs. actual 0.905), attributable to the severe 90:10 class imbalance.

2.3.3 Clustering Results. K-Means ($k = 10$) achieved a silhouette score of 0.014 and Davies-Bouldin index of 5.356 on the full 56,213-document corpus. Despite modest scores (expected for noisy, high-dimensional text data), the clusters revealed psychologically coherent themes through top-term analysis: relationship conflict, anxiety/panic, depression/hopelessness, sleep/somatic complaints, and academic/work stress. Weighted cluster purity reached 0.901, indicating strong alignment with ground-truth stress labels.

Figure 6 shows the 2D SVD projections of K-Means clusters alongside ground-truth labels, revealing partial but meaningful structural alignment.

Hierarchical Agglomerative Clustering (Ward linkage, $k = 5$) outperformed K-Means on both metrics (silhouette 0.039, DB-index 3.254; see Table 6). Subreddit×cluster heatmaps (Figure 7) confirmed community-specific concentration: r/anxiety mapped to anxiety-dominant clusters, r/depression to hopelessness clusters, and r/ptsd to trauma-specific clusters.

2.3.4 Frequent Pattern Mining Results. FP-Growth identified 1,203 frequent itemsets in stressed posts (507 bigrams, 402 trigrams) at 10% minimum support. The top-3 bigrams were {im, like} (26.7%), {im, dont} (25.9%), and {feel, like} (24.9%)—all first-person emotional expressions. Not-stressed posts yielded only 138 bigrams at a 5× lower support threshold, confirming that word co-occurrence is itself a stress-discriminative signal.

Association rules with lift >3.0 revealed patterns of emotionally uncertain self-narration: {feel, know, im} → {like, dont} (lift = 3.05) captures self-questioning language that is 3× more likely in stressed posts. The two-phase pipeline (Table 7) showed FPM features alone

Table 7: FP-Growth two-phase classification results.

Phase	Features	AUC	Macro F1
Phase 1 (FPM only)	53	0.850	0.553
Phase 2 (FPM+TF-IDF)	1,053	**0.968**	**0.786**

achieve AUC 0.850, and when combined with TF-IDF, performance improved to AUC 0.968—a 42% improvement in macro F1.

2.3.5 Cross-Model Convergence (Q2). A key finding is that *all four model categories independently identify the same stress language patterns*: Naïve Bayes log-probabilities, Decision Tree feature importance, Logistic Regression coefficients, K-Means cluster centroids, and FP-Growth itemsets converge on five categories:

(1) **Emotional distress**: anxiety, panic, depressed, hopeless, im terrified, im scared, suicidal thoughts
(2) **Physical symptoms**: tired, sleep, insomnia, exhausted, pain, headache
(3) **Relationship language**: boyfriend, breakup, family, relationship, cheating
(4) **Self-referential**: I/me/my pronoun ratio 2.1× higher in stressed posts
(5) **Help-seeking**: therapy, advice, anyone, please, need, diagnosed

This multi-model agreement—across supervised, unsupervised, and pattern mining approaches—validates these as genuine stress signals rather than artifacts of any single algorithm.

3 Conclusion & Future Work

3.1 Summary

This study demonstrates that psychological stress can be detected from Reddit text with moderate-to-high confidence using standard data mining techniques. Linear SVM achieved the best overall classification performance (F1 = 0.868, AUC = 0.883) on a 5,017-dimensional feature space combining TF-IDF and engineered sentiment/emotion features. Naïve Bayes provided the highest recall (0.910), making it suitable for applications where missing stressed individuals carries higher cost than false alarms. Clustering revealed that stress is not monolithic—it manifests in distinct thematic sub-types (anxiety, depression, relationship conflict, somatic complaints) that align with subreddit community boundaries. FP-Growth confirmed that first-person emotional co-occurrence patterns ({im, dont}, {feel, like}) are themselves predictive features, with the combined FPM+TF-IDF pipeline achieving AUC 0.968.

The most significant cross-cutting finding is the convergence of all four model categories on the same stress language patterns: emotional distress vocabulary, physical symptom mentions, self-referential language, and help-seeking behavior. This convergence across supervised, unsupervised, and pattern mining approaches provides strong evidence that these patterns reflect genuine stress expression rather than algorithmic artifacts.

3.2 Limitations

Several limitations constrain the generalizability of these findings.

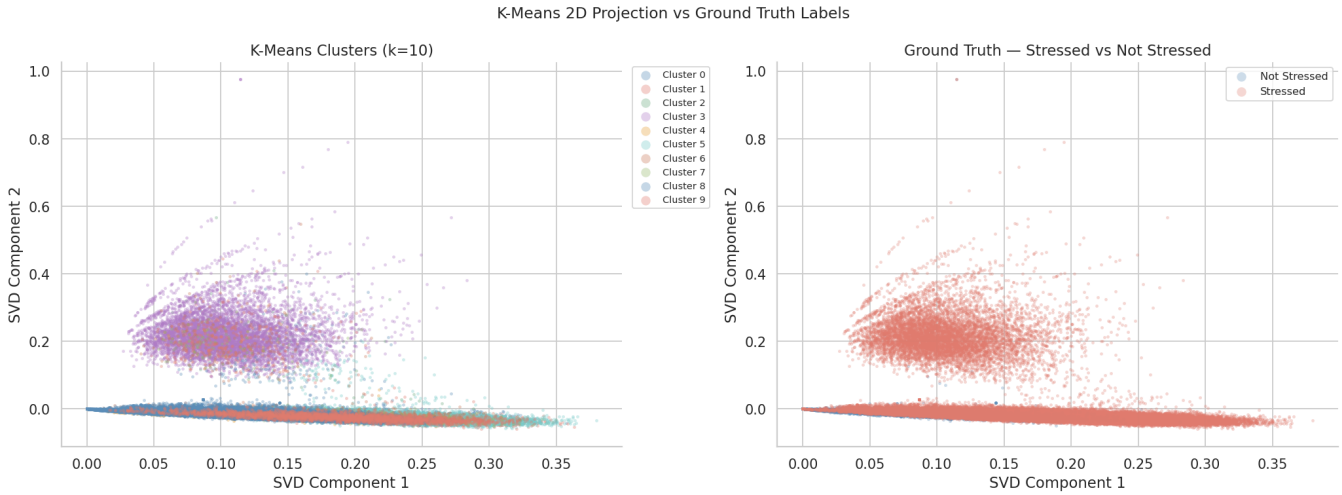

Figure 6: 2D scatter plots (SVD projection). Left: K-Means cluster assignments ($k = 10$). Right: Ground-truth stress labels. Clusters capture thematic structure that partially aligns with the binary stress labels.

Figure 7: Subreddit–cluster heatmap showing strong community-specific concentration, confirming that different stress types map to distinct thematic groups.

Platform bias: All data comes from Reddit, which skews toward younger, English-speaking, technologically literate users. Stress expression on other platforms may differ substantially.

Label noise: Subreddit-based labeling introduces systematic noise—not every post in r/anxiety reflects genuine stress. Some are informational, supportive, or recovery-focused.

Binary oversimplification: The stressed/not-stressed binary label collapses a psychological spectrum. Our confidence analysis shows 15.6% of predictions fall in the uncertain zone, precisely where a finer-grained scale would be most valuable.

Context blindness: TF-IDF and bag-of-words models cannot capture sarcasm, irony, or contextual tone—common on Reddit. Error analysis confirmed these as systematic failure modes.

Class imbalance: The regression dataset is 90% stressed posts, biasing models toward over-prediction. Calibration analysis showed Logistic Regression underestimates stress probability by 5.1 percentage points despite class weighting.

3.3 Ethical Considerations

The work we describe sits in a research tradition that carries non-trivial ethical weight, and we want to record the concerns that shaped our design decisions and that should constrain any future attempt to operationalize the techniques reported here.

Consent and expectation: Reddit posts are technically public, but users who write in r/anxiety or r/depression rarely expect their text to be ingested into training corpora for a stress classifier, and the pseudonymity that makes the platform useful for self-disclosure also means that individuals cannot meaningfully opt out of analyses they are not aware of. We use only pre-published, de-identified research datasets (Dreaddit, SWMH, HuggingLearners) and do not scrape or re-identify individual users. Any extension of this work to new Reddit data should follow the same constraint.

Harm from false positives: The cost asymmetry between false positives and false negatives is not symmetric in a mental health setting. A person falsely flagged as stressed who is then targeted with unsolicited intervention, platform warnings, or content filtering can experience real harm, particularly if the intervention is visible to others. Chancellor and De Choudhury [8] specifically caution against assuming that higher recall is always better in this domain. Our reported Naïve Bayes recall of 0.910 should be read in that light: it is useful as a research finding, not as a target for real-world deployment without careful threshold calibration.

Construct validity: A classifier trained on community-based labels learns to recognize *posts that resemble mental-health-subreddit content*, which is not the same thing as learning to recognize *clinical*

stress. Treating the two as interchangeable risks pathologizing ordinary expression and under-counting distress in users who do not post to the relevant communities. We treat our models as research instruments for studying stress language, not as screening tools.

Downstream propagation: As discussed in Section 1.1, Reddit content propagates through search engines and large language models. A stress classifier deployed at scale could influence which Reddit posts are surfaced, demoted, or rewritten by downstream systems, with consequences that extend beyond the immediate user base. Any such deployment should be accompanied by public documentation of the model's decisions, its error modes, and a mechanism for individuals to contest classifications applied to their content.

3.4 Future Work

Transformer models: BERT or RoBERTa fine-tuned on Reddit mental health data [3] could capture context and sarcasm that bag-of-words models miss.

Ordinal labels: Replacing binary labels with a multi-level scale (none/mild/moderate/severe) would enable genuine stress intensity estimation.

Longitudinal analysis: Tracking individual users across posts over time could reveal stress trajectories, escalation patterns, and crisis onset indicators, directly addressing proposal questions P3, P6, and P9 which could not be answered with the post-level supervision used in the current study.

Multi-platform validation: Testing models on Twitter, clinical datasets, or crisis helpline transcripts would assess generalizability beyond Reddit's specific culture.

Subreddit-specific models: Our per-subreddit analysis revealed significant accuracy variation; community-specific models or domain adaptation could improve performance for ambiguous communities.

Ethical deployment: Any real-world deployment of stress detection must address consent, false positive harm, and algorithmic bias in mental health contexts [8].

References

[1] C. Hutto and E. Gilbert. 2014. VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. In *Proceedings of the AAAI Conference on Weblogs and Social Media (ICWSM)*.

[2] E. Turcan and K. McKeown. 2019. Dreaddit: A Reddit Dataset for Stress Analysis in Social Media. In *Proceedings of the ACL Workshop on Computational Linguistics and Clinical Psychology (CLPsych)*.

[3] S. Ji, T. Zhang, L. Ansari, J. Fu, P. Tiwari, and E. Cambria. 2022. MentalBERT: Publicly Available Pretrained Language Models for Mental Healthcare. In *Proceedings of the Language Resources and Evaluation Conference (LREC)*.

[4] R. Agrawal and R. Srikant. 1994. Fast Algorithms for Mining Association Rules. In *Proceedings of the International Conference on Very Large Data Bases (VLDB)*. 487–499.

[5] J. Han, J. Pei, and Y. Yin. 2004. Mining Frequent Patterns without Candidate Generation: A Frequent-Pattern Tree Approach. *Data Mining and Knowledge Discovery* 8, 1 (2004), 53–87.

[6] F. Pedregosa et al. 2011. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12 (2011), 2825–2830.

[7] G. Coppersmith, R. Dredze, and C. Harman. 2018. Natural Language Processing of Social Media as Screening for Suicide Risk. *Biomedical Informatics Insights* 10 (2018).

[8] S. Chancellor and M. De Choudhury. 2020. Methods in Predictive Techniques for Mental Health Status on Social Media: A Critical Review. *NPJ Digital Medicine* 3, 43 (2020).